A Ablation Study with Memory Constraints

Backprop Len	Recurrence	Encoding	Loss	pplx best	pplx init	Attn Len
128	✓	Ours	Full	**26.77**	**27.02**	**500**
128	✓	Ours	Partial	28.33	28.69	460
176	✗	Ours	Full	27.98	28.43	400
172	✗	Ours	Partial	28.83	28.83	120

Table 10: Ablation study on WikiText-103 with the same GPU memory constraints.

Table 10 compares Transformer-XL with baseline under the same memory budget. Transformer-XL still outperforms the baseline even with a shorter backprop length.

B Efficient Computation of the Attention with Relative Positional Embedding

As we discussed in section 3.3, the naive way of computing the $\mathbf{W}_{k,R}\mathbf{R}_{i-j}$ for all pairs (i, j) is subject to a quadratic cost. Here, we present a simple method with only a linear cost. Firstly, notice that the relative distance $i - j$ can only be integer from 0 to $M + L - 1$, where M and L are the memory length and segment length respectively. Hence, the rows of the matrix

$$\mathbf{Q} := \begin{bmatrix} \mathbf{R}_{M+L-1}^\top \\ \mathbf{R}_{M+L-2}^\top \\ \vdots \\ \mathbf{R}_1^\top \\ \mathbf{R}_0^\top \end{bmatrix} \mathbf{W}_{k,R}{}^\top = \begin{bmatrix} [\mathbf{W}_{k,R}\mathbf{R}_{M+L-1}]^\top \\ [\mathbf{W}_{k,R}\mathbf{R}_{M+L-2}]^\top \\ \vdots \\ [\mathbf{W}_{k,R}\mathbf{R}_1]^\top \\ [\mathbf{W}_{k,R}\mathbf{R}_0]^\top \end{bmatrix} \in \mathbb{R}^{(M+L)\times d}$$

consist of all possible vector outputs of $\mathbf{W}_{k,R}\mathbf{R}_{i-j}$ for any (i, j). Note that we have defined $\mathbf{Q}$ in a reversed order, i.e., $\mathbf{Q}_k = \mathbf{W}_{k,R}\mathbf{R}_{M+L-1-k}$, to make further discussion easier.

Next, we collect the term (b) for all possible i, j into the following $L \times (M + L)$ matrix,

$$\begin{aligned} \mathbf{B} &= \begin{bmatrix} q_0^\top \mathbf{W}_{k,R}\mathbf{R}_M & \cdots & q_0^\top \mathbf{W}_{k,R}\mathbf{R}_0 & 0 & \cdots & 0 \\ q_1^\top \mathbf{W}_{k,R}\mathbf{R}_{M+1} & \cdots & q_1^\top \mathbf{W}_{k,R}\mathbf{R}_1 & q_1^\top \mathbf{W}_{k,R}\mathbf{R}_0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ q_{L-1}^\top \mathbf{W}_{k,R}\mathbf{R}_{M+L-1} & \cdots & q_{L-1}^\top \mathbf{W}_{k,R}\mathbf{R}_{M+L-1} & q_{L-1}^\top \mathbf{W}_{k,R}\mathbf{R}_{L-1} & \cdots & q_{L-1}^\top \mathbf{W}_{k,R}\mathbf{R}_0 \end{bmatrix} \\ &= \begin{bmatrix} q_0^\top \mathbf{Q}_{L-1} & \cdots & q_0^\top \mathbf{Q}_{M+L-1} & 0 & \cdots & 0 \\ q_1^\top \mathbf{Q}_{L-2} & \cdots & q_1^\top \mathbf{Q}_{M+L-2} & q_1^\top \mathbf{Q}_{M+L-1} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ q_{L-1}^\top \mathbf{Q}_0 & \cdots & q_{L-1}^\top \mathbf{Q}_M & q_{L-1}^\top \mathbf{Q}_{M+1} & \cdots & q_{L-1}^\top \mathbf{Q}_{M+L-1} \end{bmatrix} \end{aligned}$$

Then, we further define

$$\widetilde{\mathbf{B}} = \mathbf{q}\mathbf{Q}^\top = \begin{bmatrix} q_0^\top \mathbf{Q}_0 & \cdots & q_0^\top \mathbf{Q}_M & q_0^\top \mathbf{Q}_{M+1} & \cdots & q_0^\top \mathbf{Q}_{M+L-1} \\ q_1^\top \mathbf{Q}_0 & \cdots & q_1^\top \mathbf{Q}_M & q_1^\top \mathbf{Q}_{M+1} & \cdots & q_1^\top \mathbf{Q}_{M+L-1} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ q_{L-1}^\top \mathbf{Q}_0 & \cdots & q_{L-1}^\top \mathbf{Q}_M & q_{L-1}^\top \mathbf{Q}_{M+1} & \cdots & q_{L-1}^\top \mathbf{Q}_{M+L-1} \end{bmatrix}.$$

Now, it is easy to see an immediate relationship between $\mathbf{B}$ and $\widetilde{\mathbf{B}}$, where the i-th row of $\mathbf{B}$ is simply a left-shifted version of i-th row of $\widetilde{\mathbf{B}}$. Hence, the computation of $\mathbf{B}$ only requires a matrix multiplication $\mathbf{q}\mathbf{Q}^\top$ to compute $\widetilde{\mathbf{B}}$ and then a set of left-shifts.

Similarly, we can collect all term (d) for all possible i, j into another $L \times (M + L)$ matrix $\mathbf{D}$,

$$\mathbf{D} = \begin{bmatrix} v^\top \mathbf{Q}_{L-1} & \cdots & v^\top \mathbf{Q}_{M+L-1} & 0 & \cdots & 0 \\ v^\top \mathbf{Q}_{L-2} & \cdots & v^\top \mathbf{Q}_{M+L-2} & v^\top \mathbf{Q}_{M+L-1} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ v^\top \mathbf{Q}_0 & \cdots & v^\top \mathbf{Q}_M & v^\top \mathbf{Q}_{M+1} & \cdots & v^\top \mathbf{Q}_{M+L-1} \end{bmatrix}.$$